Known-source drop: point mode

UMA-8 + ODAS + MobileNetV2 on Jetson — changes made to drop the known source correctly

1. Why the previous method failed

The 15 s null calibration learned the known source's **sound** (spatial covariance per frequency: directions, energy, spectrum) during those 15 s. The known source is a video whose sound keeps changing, so frequencies and levels not seen during calibration were no longer suppressed and reached MobileNet.

2. Changes

1. **No null calibration, one ODAS.** The null feeder, the second (nulled) ODAS, the fusion and the old calibration were removed. `arecord` writes the Loopback audio into `/tmp/odas_raw.fifo`, read by ONE ODAS (`uma-8/config/uma8.cfg`) that publishes `/raw/sst` and `/raw/sss`. Nothing depends on what the known source plays.
2. **The known source is calibrated as a point** (`nodes/direction_gate_node.py`, 10 s, known source alone): the most active track is the known source → its **track id**; its **direction** = 3D mean of its positions (folded to the upper hemisphere: the flat UMA-8 cannot tell above from below); its **cone** = 95 % of its spread + 5°, limited to 15–30°. Saved in `uma-8/config/known_point.json`.
3. **Drop before MobileNet (gate).** A track is the known source if it has the known id, or its direction is inside the cone (that new id is then blocked for its whole life). Its audio channel is **silenced** in `/gate/sss` and its track is **removed** from `/gate/sst`. MobileNet reads `/gate/*`.
4. **Separated audio in MobileNet's format.** The raw path produced 48 kHz audio, MobileNet expects 16 kHz: frames could not be paired. `uma8.cfg` now outputs the separated audio at 16 kHz, 384 samples/frame, 16 bit.
5. **Pairing matched to the classifier's rule.** MobileNet pairs the k-th listed track with the k-th channel carrying audio, and both counts must be equal. The gate first kept the known track in the list while silencing its channel → every frame mismatched. It now removes the track too, and maps tracks to channels with the same rule (ODAS lists only existing tracks). `pairing_mismatch`: from 100 % of frames to ≈ 2 % (only track births/deaths).
6. **Tools.** `uma-8/scripts/start.sh`: one-command start (`load` reuses the saved point, `stop` stops all). `uma-8/scripts/sss_rec.sh`: records MobileNet's input before and after the drop, each channel labelled with its track; `tools/sss_analyze.py` reports levels, tracks and a verdict.

3. Processing chain (point mode)

```
UMA-8 -> laptop (arecord) -> Ethernet -> Jetson Loopback
-> arecord -> /tmp/odas_raw.fifo -> ODAS: /raw/sst (tracks) + /raw/sss (4 ch, 16 kHz)
-> direction_gate_node (known point: id + direction cone)
     -> /gate/sss (known channel silenced) -> MobileNet -> /classifier/tracks
     -> /gate/sst (known track removed)    -> target_direction_node -> /target/*
-> drone_warning_node (/drone_warning: CLEAR | DRONE x y z) | RViz (/target/markers)
```

4. Results

Situation	MobileNet input	Result
Known source ON, alone	known track removed; only weak background noise passes	no drone / [CLEAR]
Known source OFF	background noise only (≈ –78 dBFS)	no drone / [CLEAR]
Room background noise	weak track (p ≈ 0.30–0.40)	rejected by MobileNet
Pairing audio ↔ tracks	pairing_mismatch ≈ 2 % (track births/deaths)	OK
Target drone (phone B)	must pass as TARGET, ≥ –65 dBFS	to be validated

5. Known limits

- Without the null, reflections of the known source (walls, ceiling) come from other directions and are not part of the point; they can reach MobileNet (so far rejected as no drone). Option: learn repeating reflection points during the same 10 s.
- If the known source moves, recalibrate (`start.sh` without `load`).
- A target inside the known cone is dropped too: keep the target ≥ 60° away from the known source.

6. Run

```
JETSON  nc -l 5005 | aplay -D hw:CARD=Loopback,DEV=0 -c 8 -r 48000 -f S32_LE -t raw ...
LAPTOP  arecord -D plughw:CARD=SPK,DEV=0 -c 8 -r 48000 -f S32_LE -t raw \
          | nc -N 192.168.10.1 5005
JETSON  cd ~/known_source_drop && uma-8/scripts/start.sh
          (known source alone for the first 10 s)
JETSON  uma-8/scripts/warning.sh                (CLEAR / DRONE x y z)
LAPTOP  rviz2 -d ~/drone_target_view.rviz   (omnisphere)
```